SeeWandel Climate

Interreg
Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union

Regime Shifts in Plankton Dynamics: What Drives Them and Can We Predict Them?

Dr. Leonardo Capitani
PI: Dr. Francesco Pomati
Department of Aquatic Ecology, Eawag · SeeWandel-Climate

www.seewandel.org

Ecomeets @ Eawag, Dübendorf
28 October 2026

Open science: take the slides with you

Scan to download the slides

SORTEE
Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology

Slides, code and data shared openly
leomarameo7.github.io/science_communication.html

Acknowledgements

SeeWandel Climate

Interreg Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union



Pomati lab · Plankton Ecology, Eawag



Steve Munch & Tanya Rogers
EDM workshop, supported by the Biodiversity Center WSL–Eawag



Prof. D. Straile



Prof. U. Gaedke



Dr. J. Alexander



Dr. P. Teiber-Sießegger

Take home message

1. Plankton regime shifts can be detected and partly explained by biotic drivers.

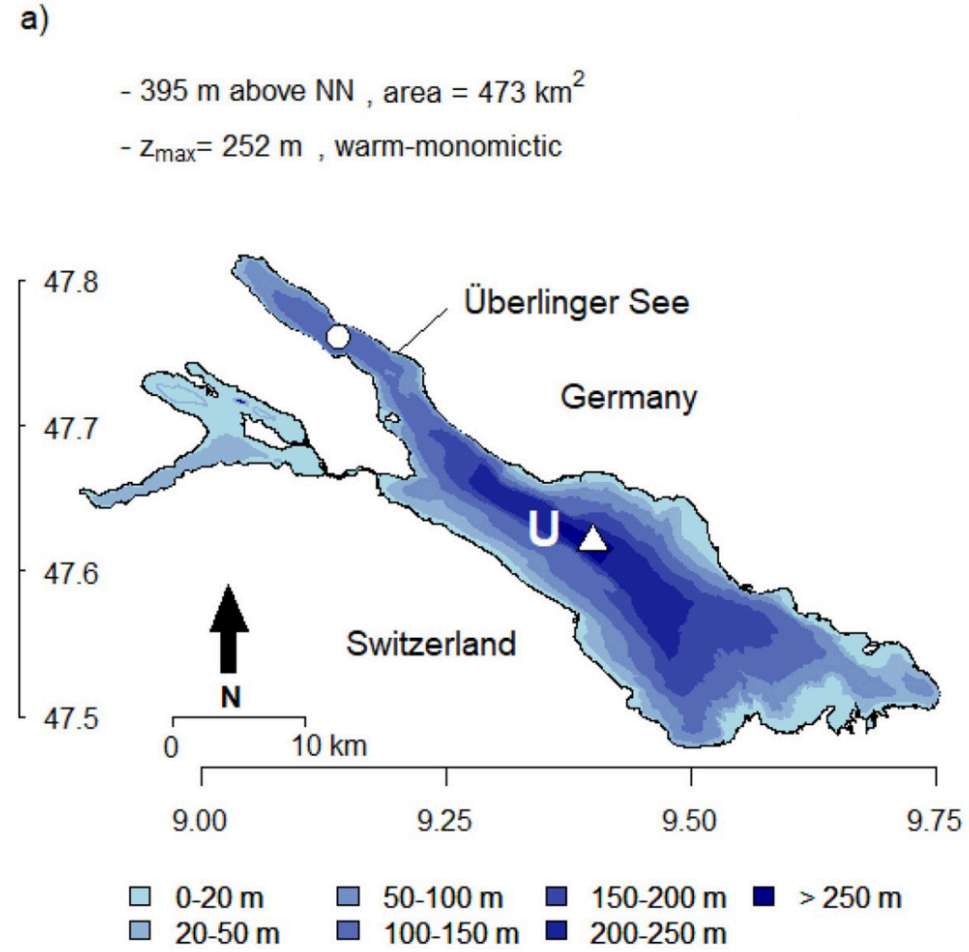
2. Whether they propagate and can be predicted is still open.

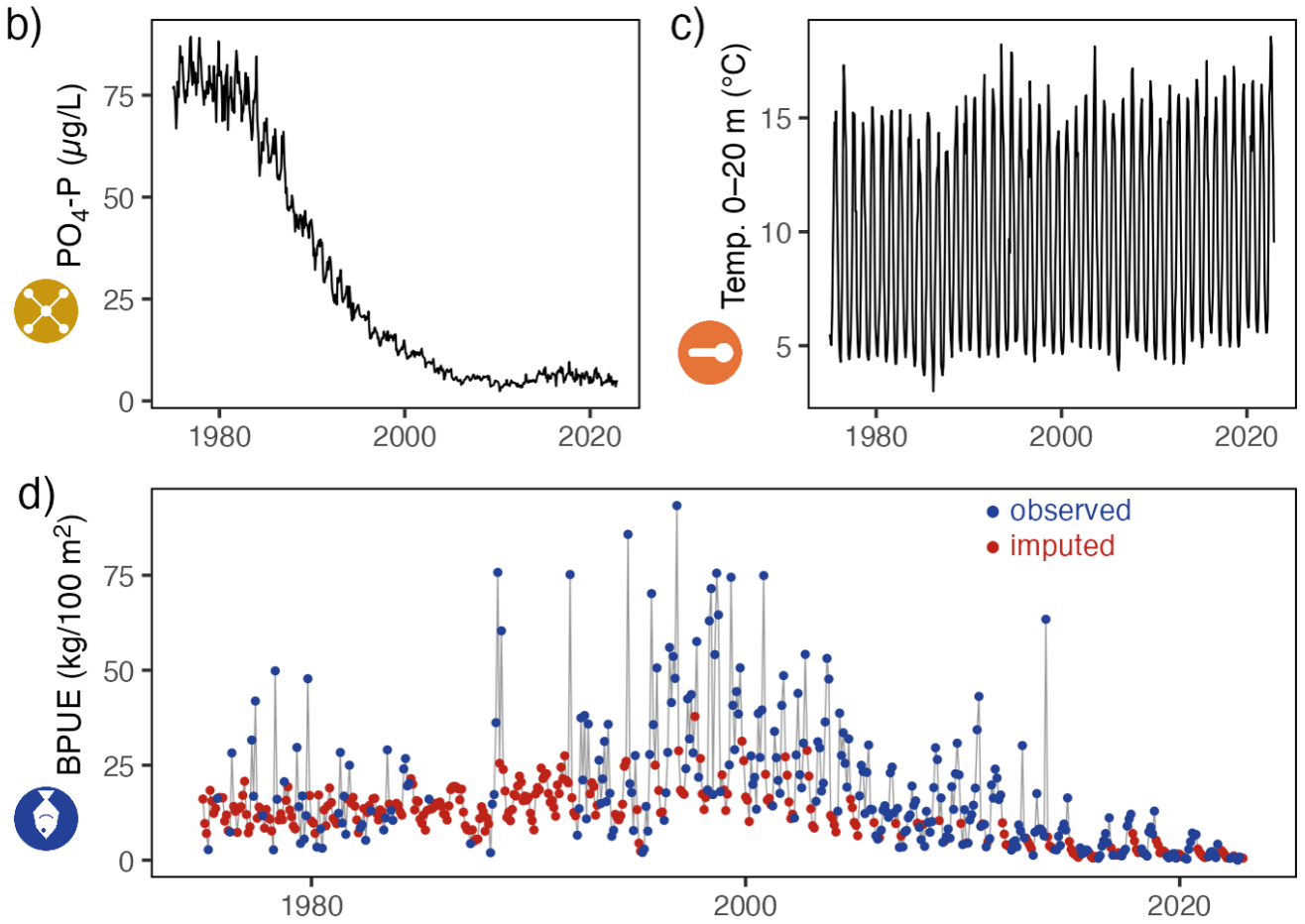


Outline

1. What we know: background
2. Research questions
3. Methods: our pipeline
4. Results: detection of regime shifts
5. Explain regime shifts
6. Propagation of regime shifts
7. Conclusions

What we know: background





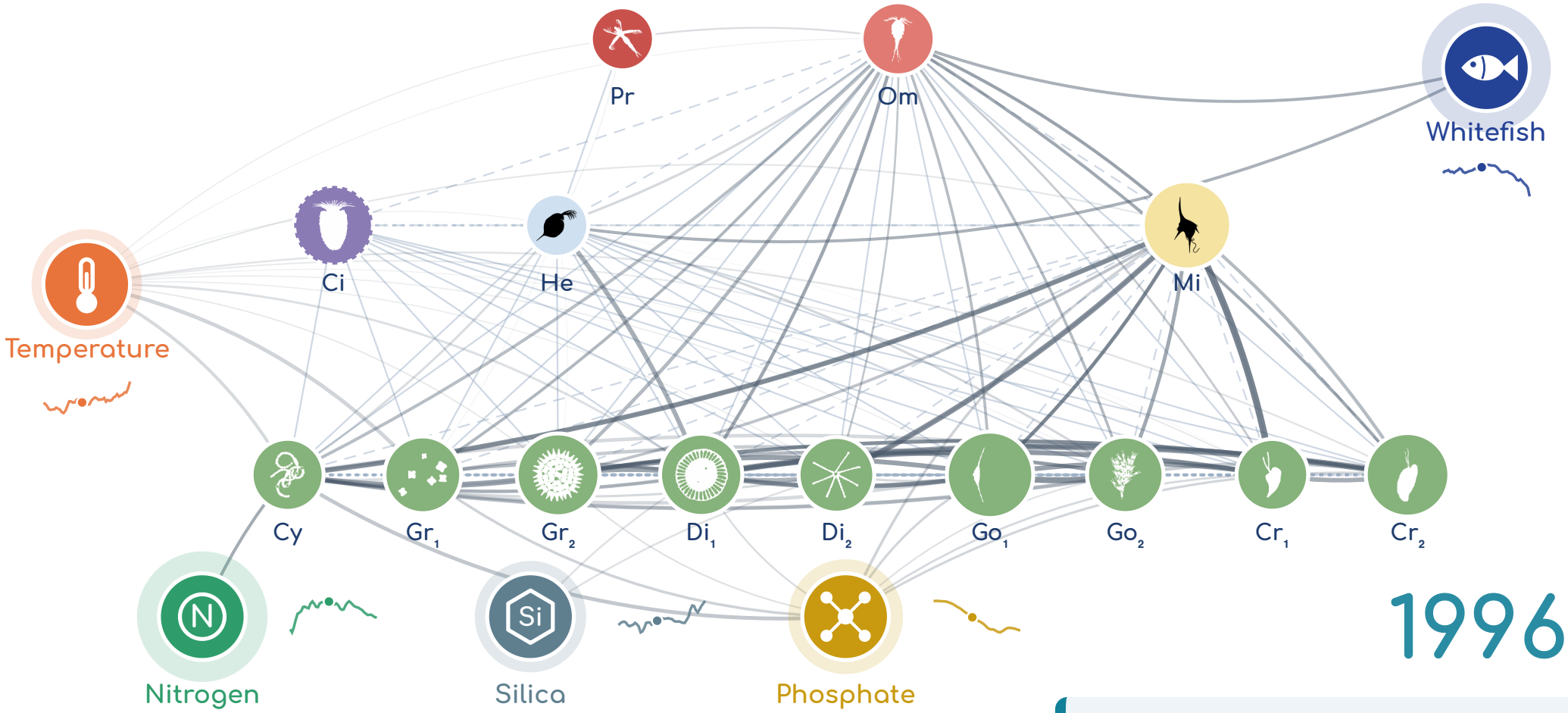
Lake Constance · 473 km²

Phosphate ↑ until 1980, then ↓

Temperature ↑ about 1 °C

Map (a): Straile 2025 · b–d: monthly IGKB and IBKF data, 1975–2022 (red: imputed)

What we know: the plankton community



Study period: **1975–2022** · monthly time series

The plankton community changes across time

Network modified from Merz et al. 2023, Nat. Clim. Change. Links without a strength: minimum intensity.

Research questions



1 · Detect

Abrupt shifts?



2 · Explain

What drives them?



3 · Propagate

Does a shift spread?



4 · Predict

Can we forecast them? (in progress)

Methods: our pipeline

1

Data

Monthly data 1975–2022

2

Detect

BEAST + two criteria

3

Explain

GP-EDM + hypotheses

4

Propagate

Shift-removal test

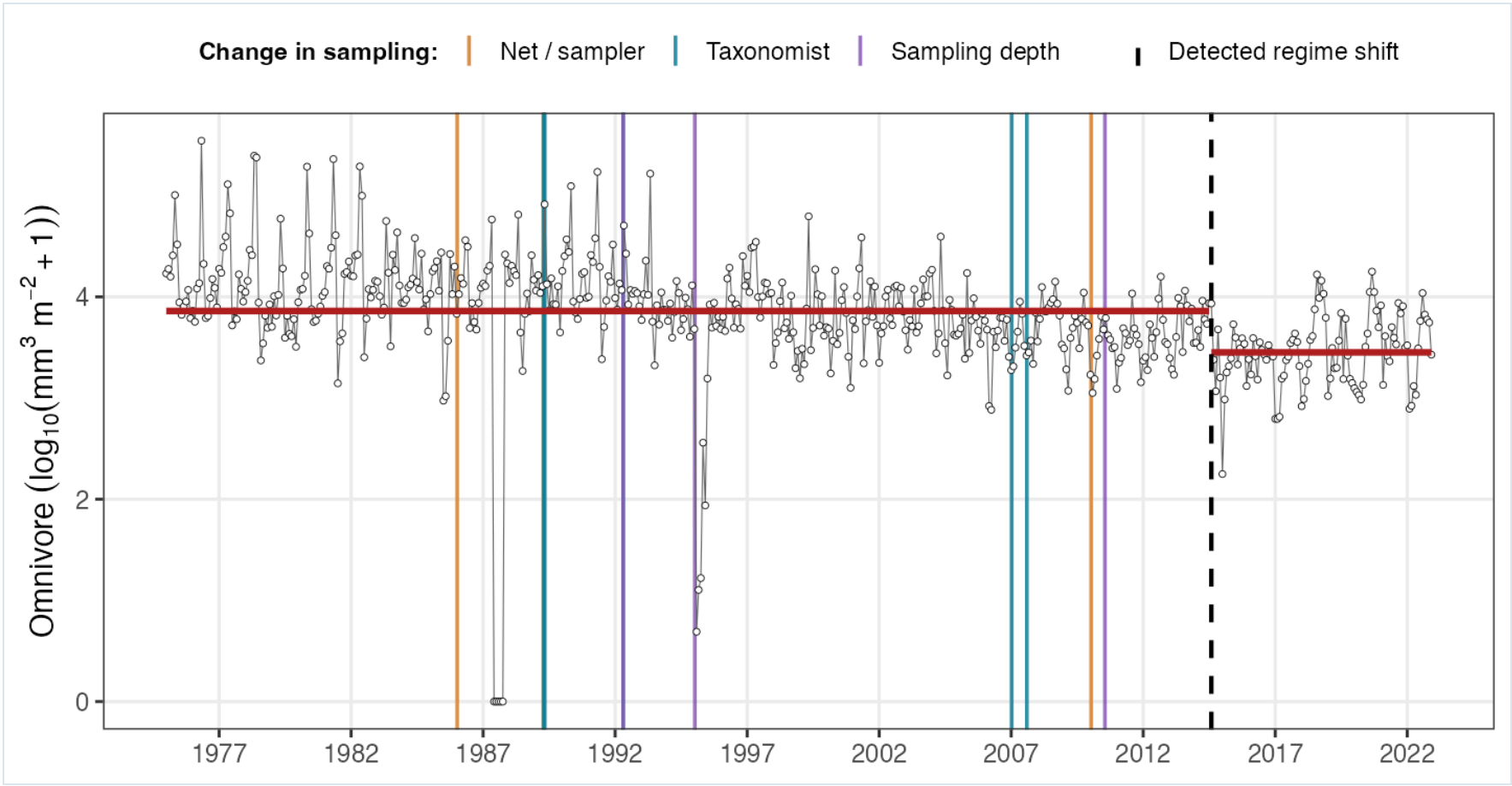
5

Predict

Forward validation

WORK IN PROGRESS

Results: detection of regime shifts





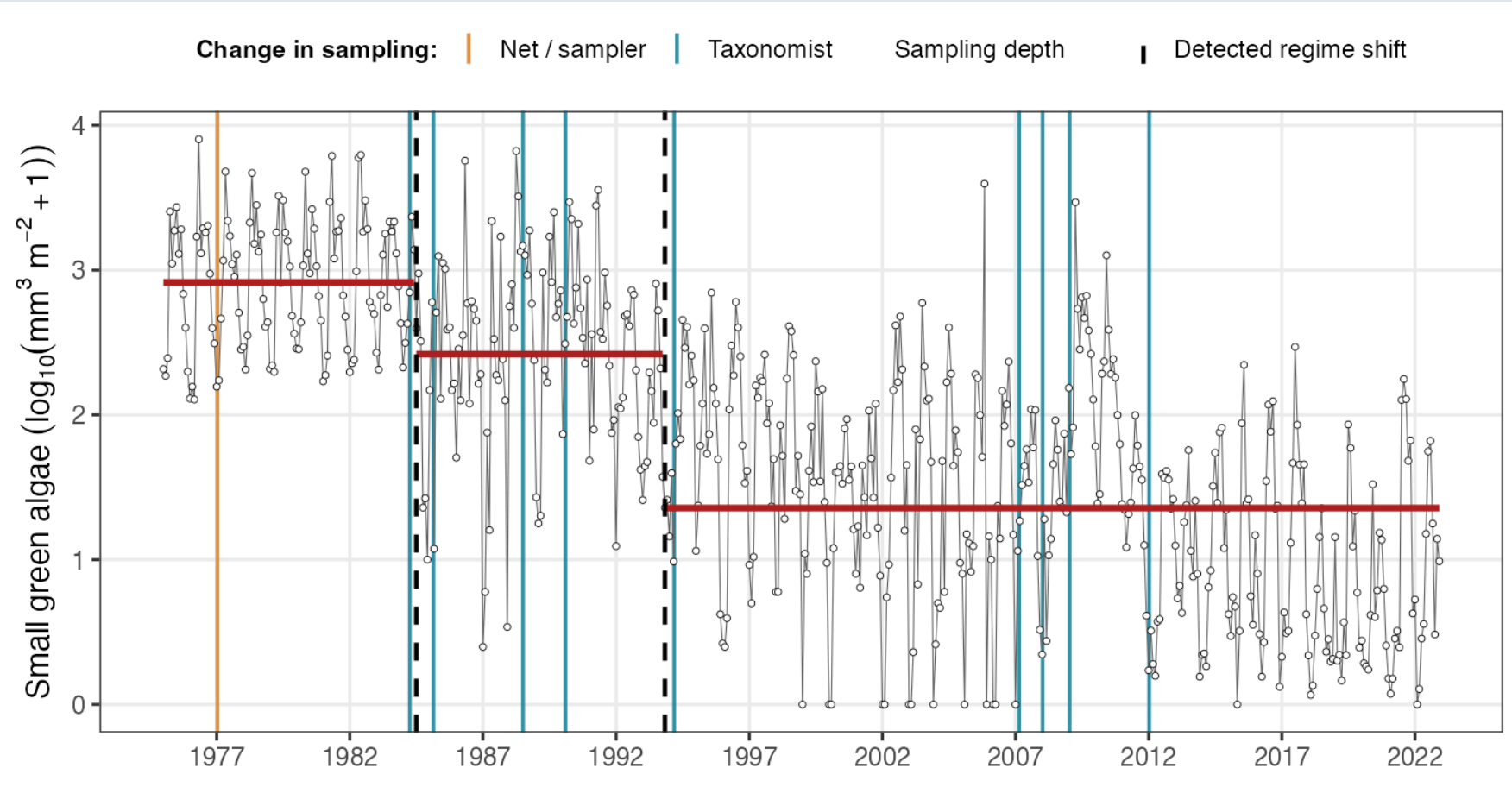
Shift: Aug 2014

Abruptness 1.19
jump size vs. usual swings (> 1 = abrupt)

Omnivores, monthly biovolume. Red: period means. The shift does not coincide with any change in sampling.

The omnivores dropped abruptly in 2014

Results: detection of regime shifts



Small green algae, monthly biovolume. Red: period means.



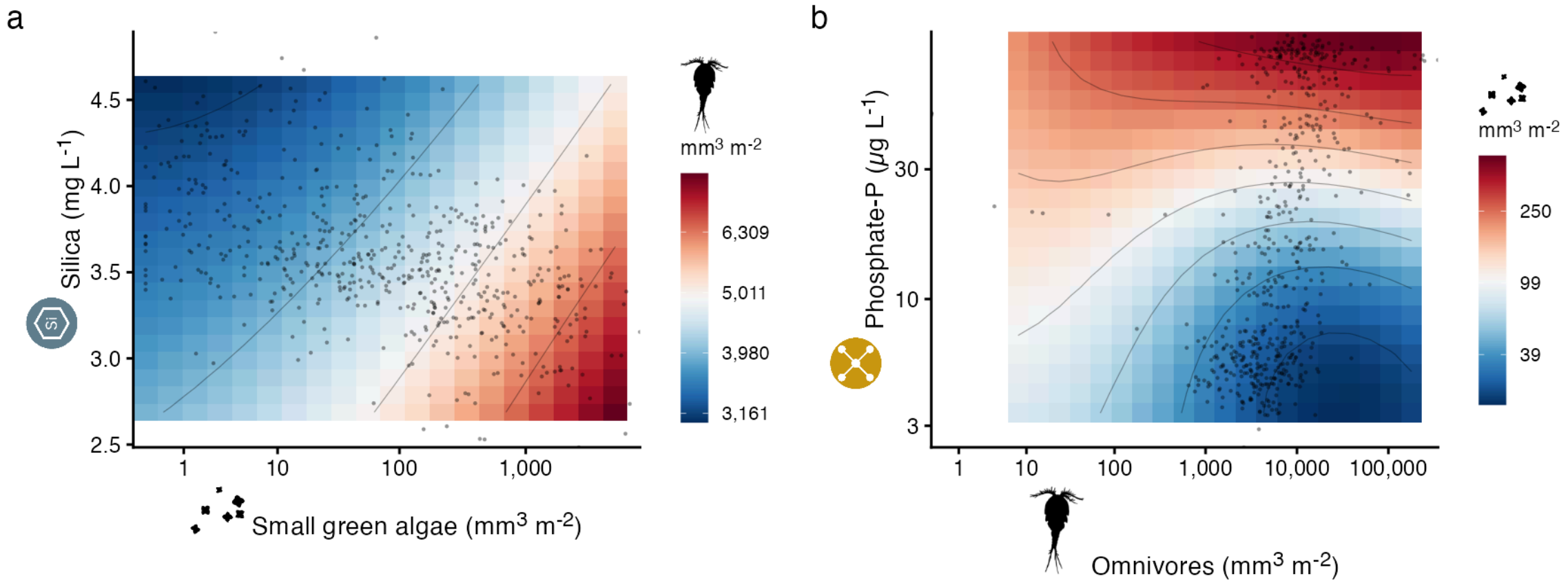
Jul 1984
abruptness **1.20**

Nov 1993
abruptness **1.15**

Both ≤ 4 months from a taxonomist change

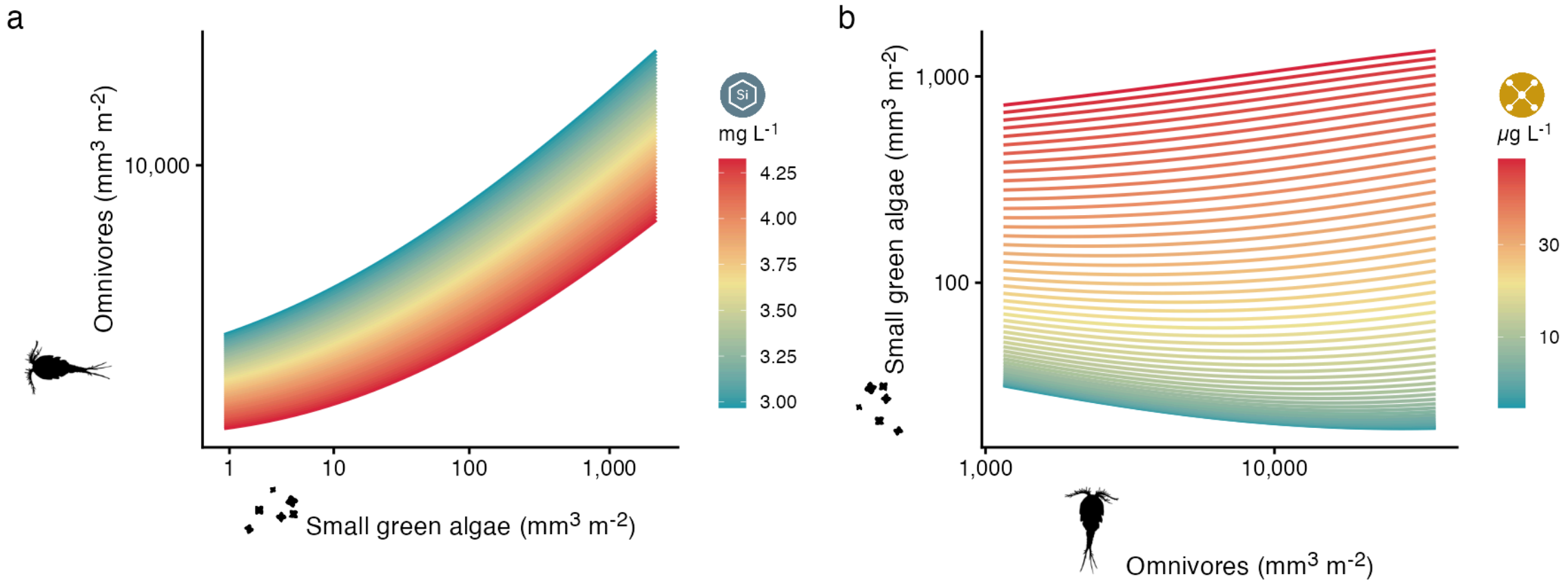
Small green algae shifted twice

Explain regime shifts



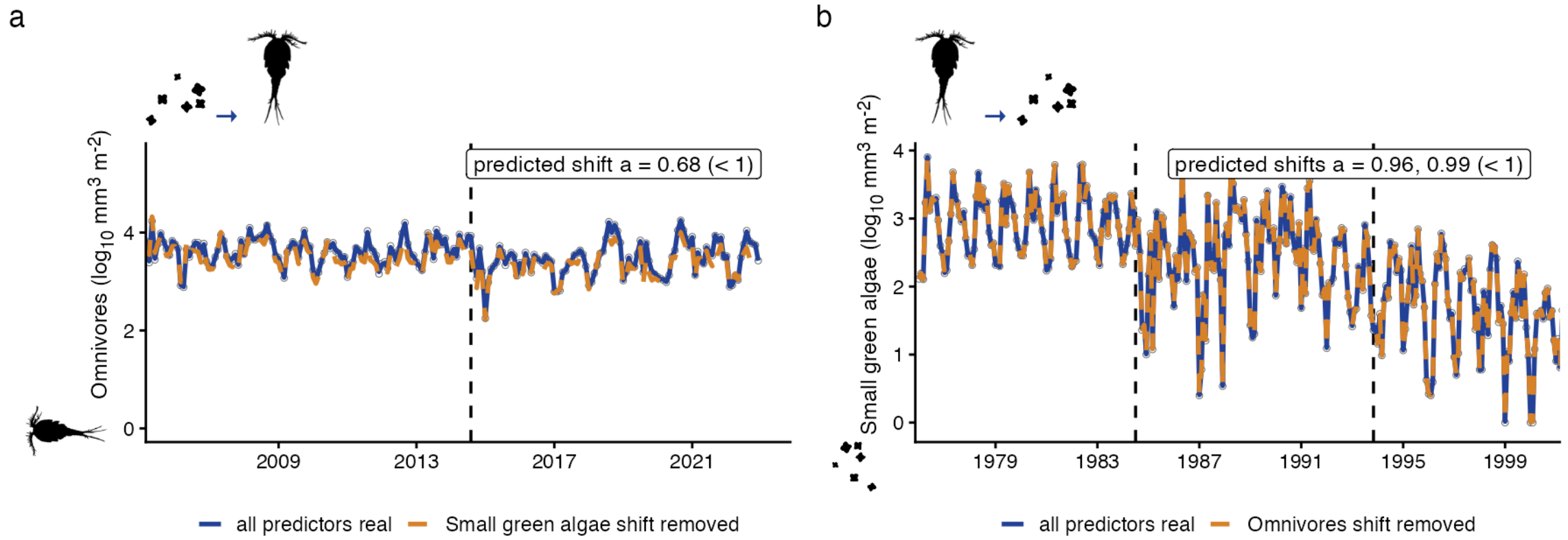
Phosphate drives small green algae

Explain regime shifts: partial dependence



Green algae favour omnivores

Results: propagation of regime shift

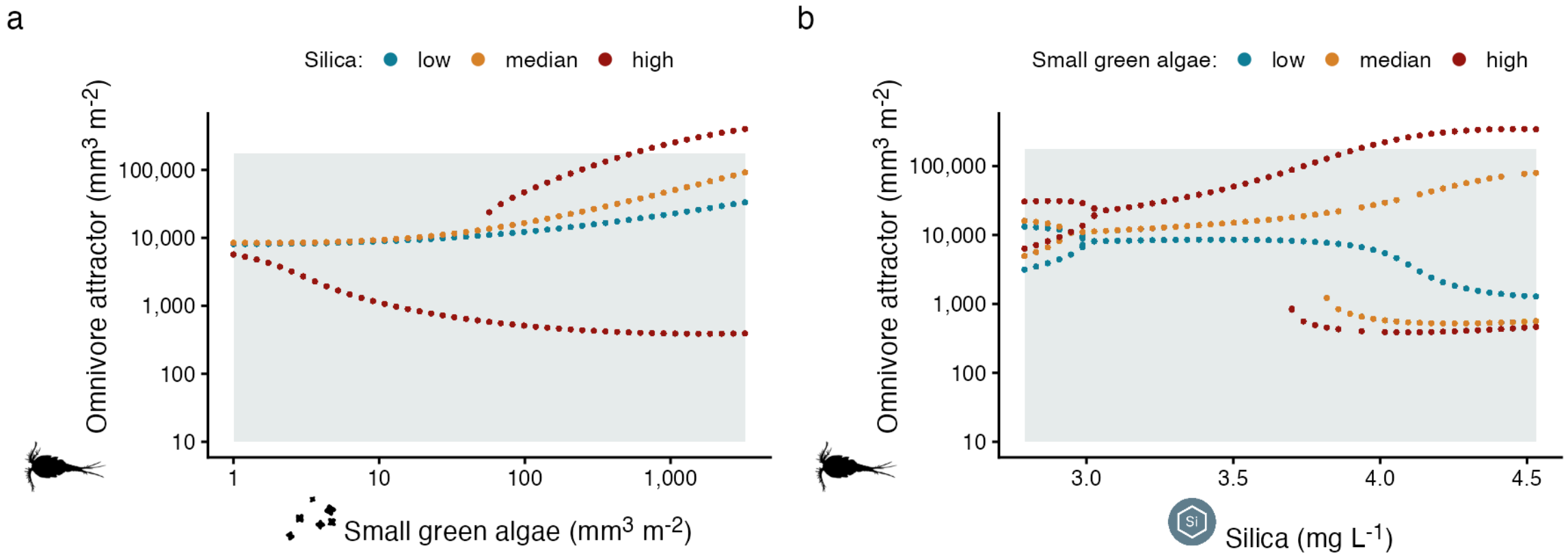


Shift-removal test: **0 of 3** shifts reproduced by the model (a < 1), so none is testable

No evidence of propagation

Regime shift: prediction

🚧 In progress · preliminary result



Two stable states at high silica

Conclusions



Shifts are detectable



Biotic drivers matter



Propagation, prediction: open

Take home message

1. Plankton regime shifts can be detected and partly explained by biotic drivers.

2. Whether they propagate and can be predicted is still open.



Acknowledgements

SeeWandel Climate

Interreg Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union

eawag aquatic research · FFS · LAZBW · isf Institut für Seenforschung · LUBW · KUP Kobus und Partner · universität innsbruck · Universität Konstanz

Bodensee-Wasserversorgung · Buffalo State Great Lakes Center · Stadt Zürich Departement der Industriellen Betriebe · igkb Internationale Gewässerschutzkommission für den Bodensee · IBKF · LMU Ludwig-Maximilians-Universität München · Baden-Württemberg Landesamt für Denkmalpflege im Regierungspräsidium Stuttgart · Internationale Rheinregulierung

Thanks to the whole **Pomati lab** group · Prof. **Steve Munch** · Dr. **Tanya Rogers** · Dr. **Luis J. Gilarranz**



Prof. D. Straile



Prof. U. Gaedke



Dr. J. Alexander

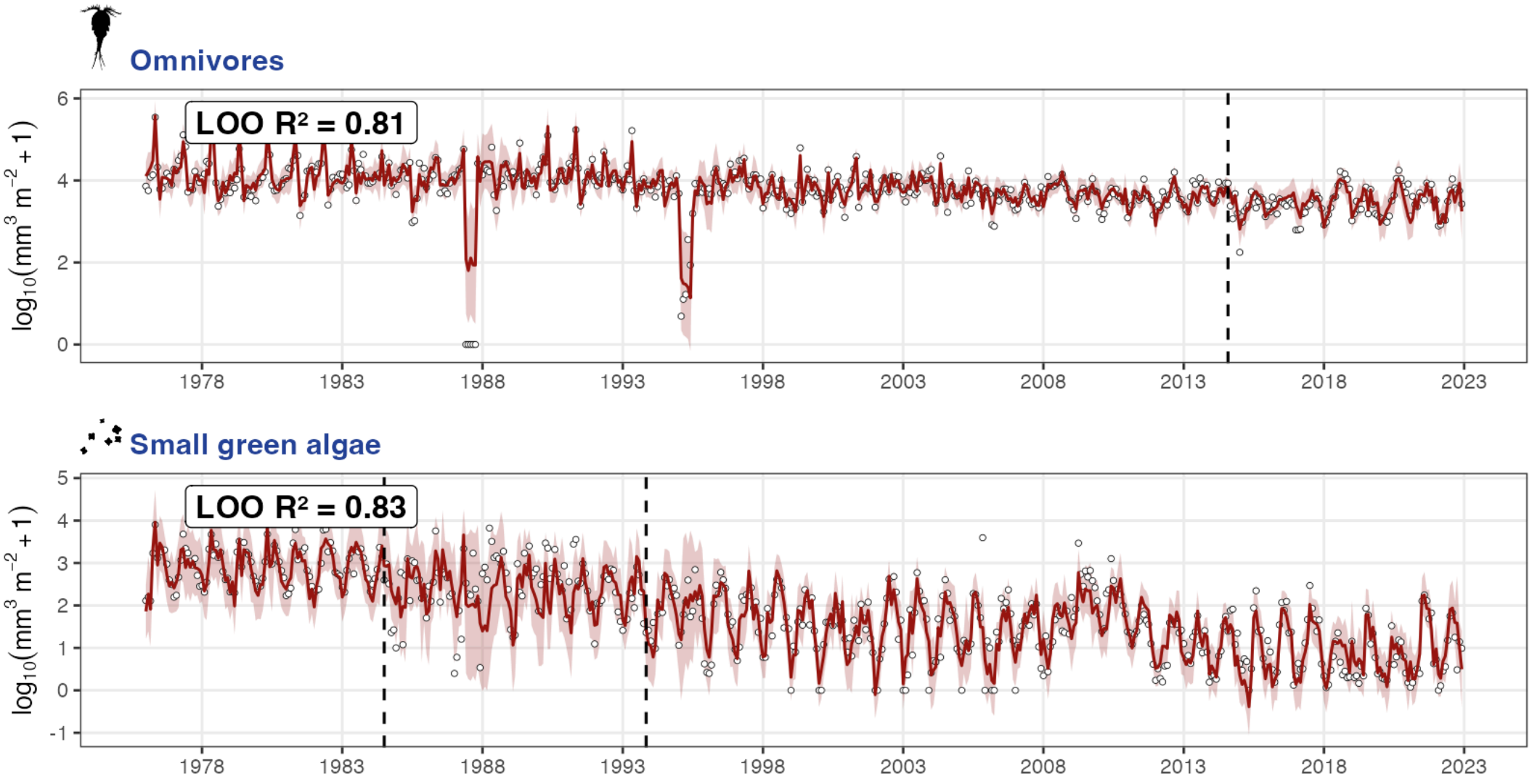


Dr. P. Teiber-Sießegger

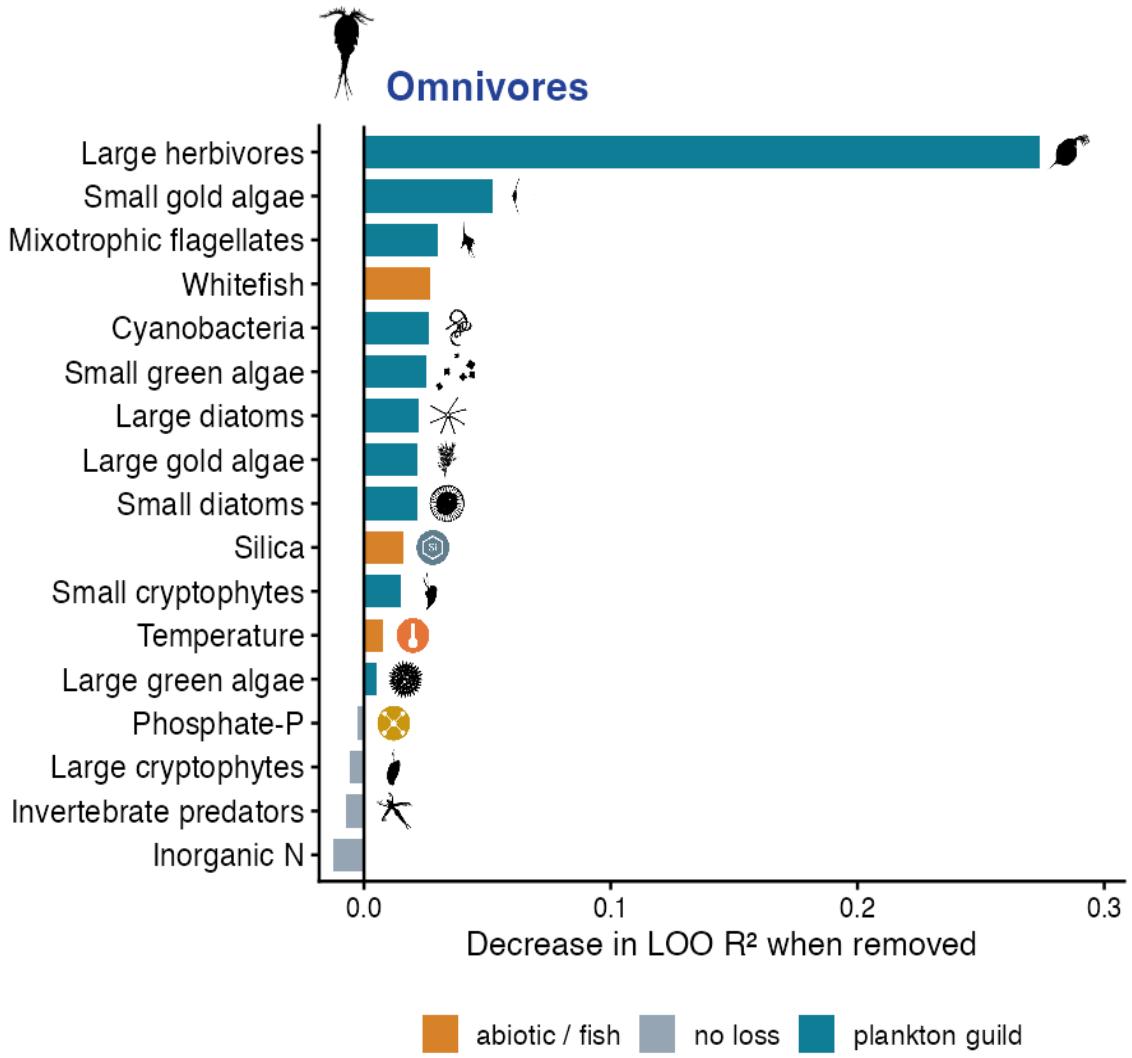
Supplementary materials

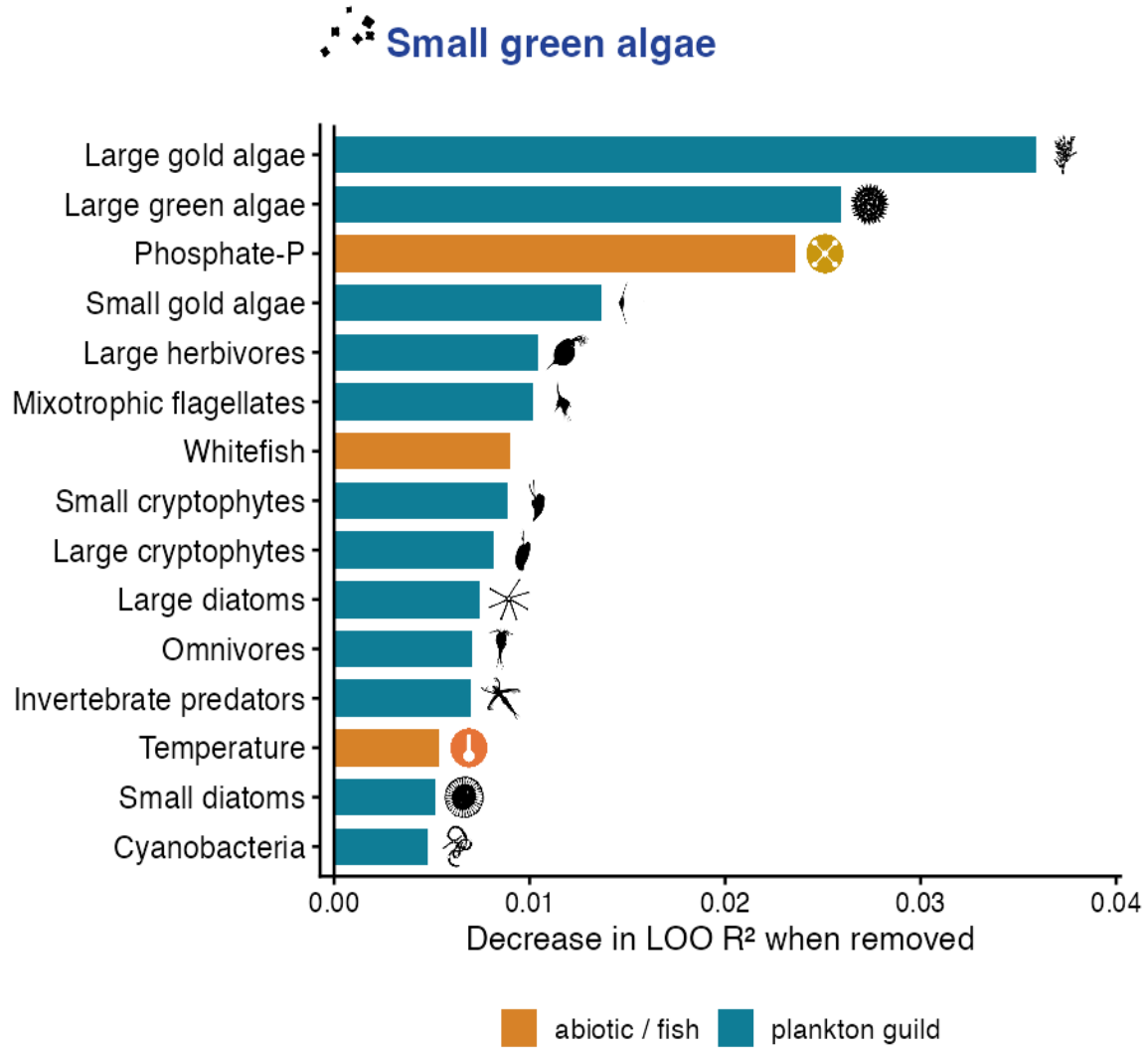


Model fit: Leave-One-Out R^2



Which predictors matter?





Which predictors were used?

Omnivores LOO R^2 = 0.81

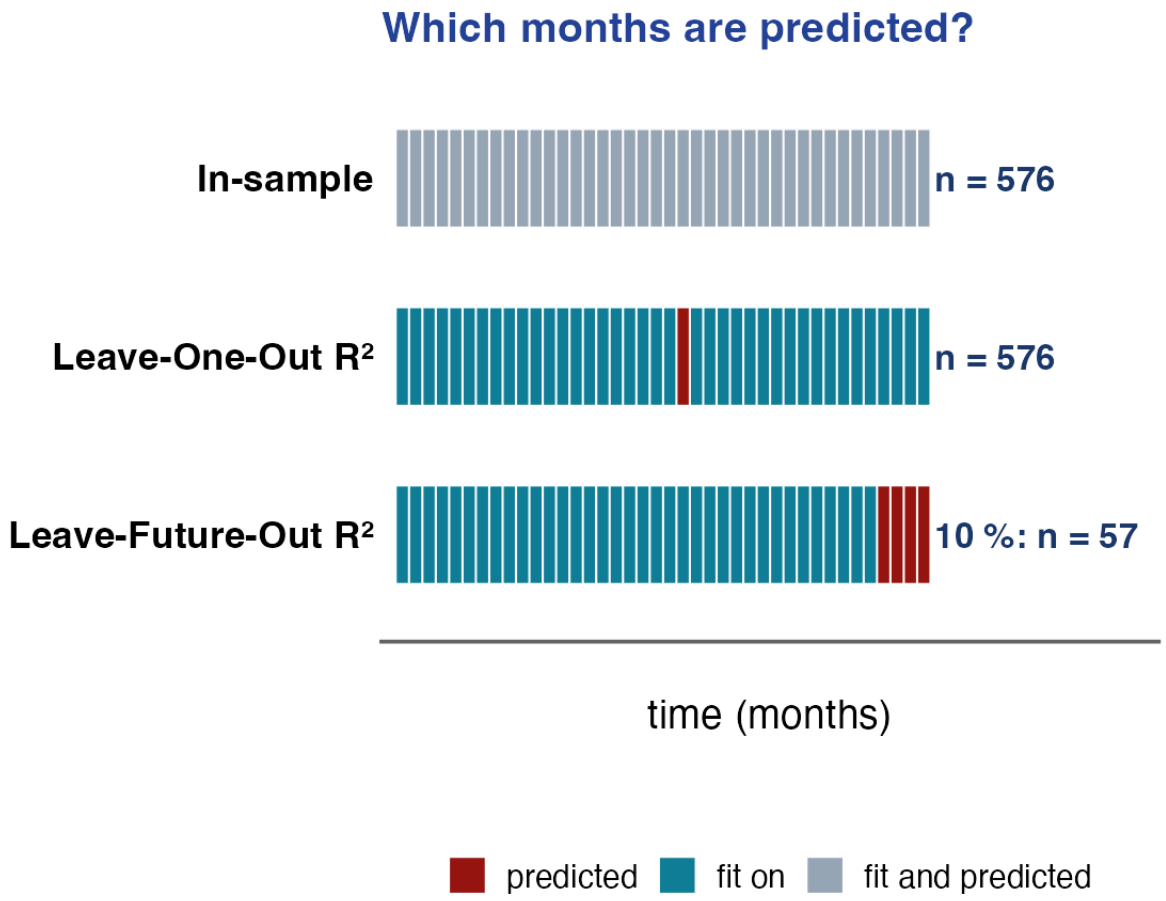
- Phosphate-P, 6 lags
- Large herbivores, 4 lags
- Whitefish, 4 lags
- Invertebrate predators, 7 lags
- Temperature, 4 lags
- Small green algae, 3 lags
- Small cryptophytes, 3 lags
- Large gold algae, 2 lags
- Small diatoms, 2 lags
- Cyanobacteria, 1 lags
- Large green algae, 2 lags
- Small gold algae, 2 lags
- Large diatoms, 2 lags
- Mixotrophic flagellates, 1 lags
- Silica, 1 lags
- Large cryptophytes removed by ARD
- Inorganic N removed by ARD

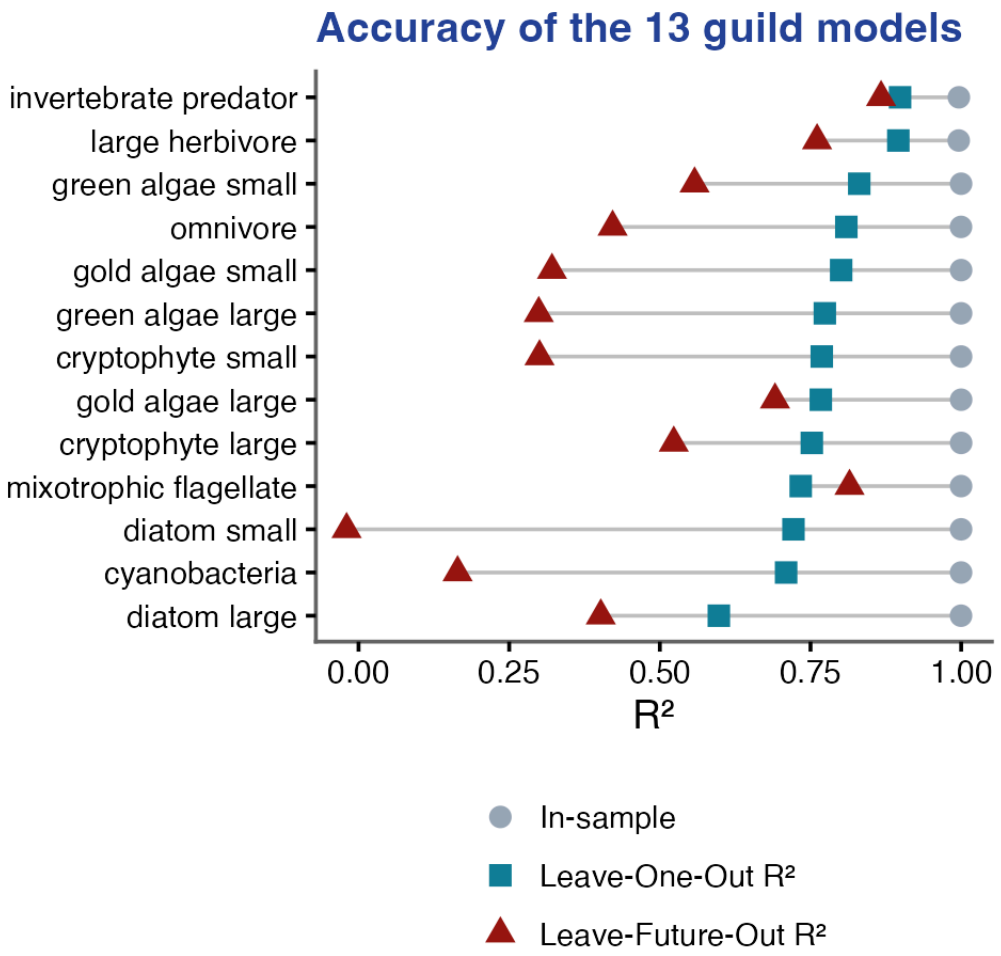
Small green algae LOO R^2 = 0.83

- Phosphate-P, 3 lags
- Large green algae, 8 lags
- Large cryptophytes, 4 lags
- Temperature, 2 lags
- Large gold algae, 4 lags
- Omnivores, 6 lags
- Large herbivores, 4 lags
- Mixotrophic flagellates, 3 lags
- Invertebrate predators, 3 lags
- Large diatoms, 3 lags
- Cyanobacteria, 4 lags
- Whitefish, 5 lags
- Small cryptophytes, 3 lags
- Small gold algae, 3 lags
- Small diatoms, 3 lags
- Inorganic N not entered
- Silica not entered

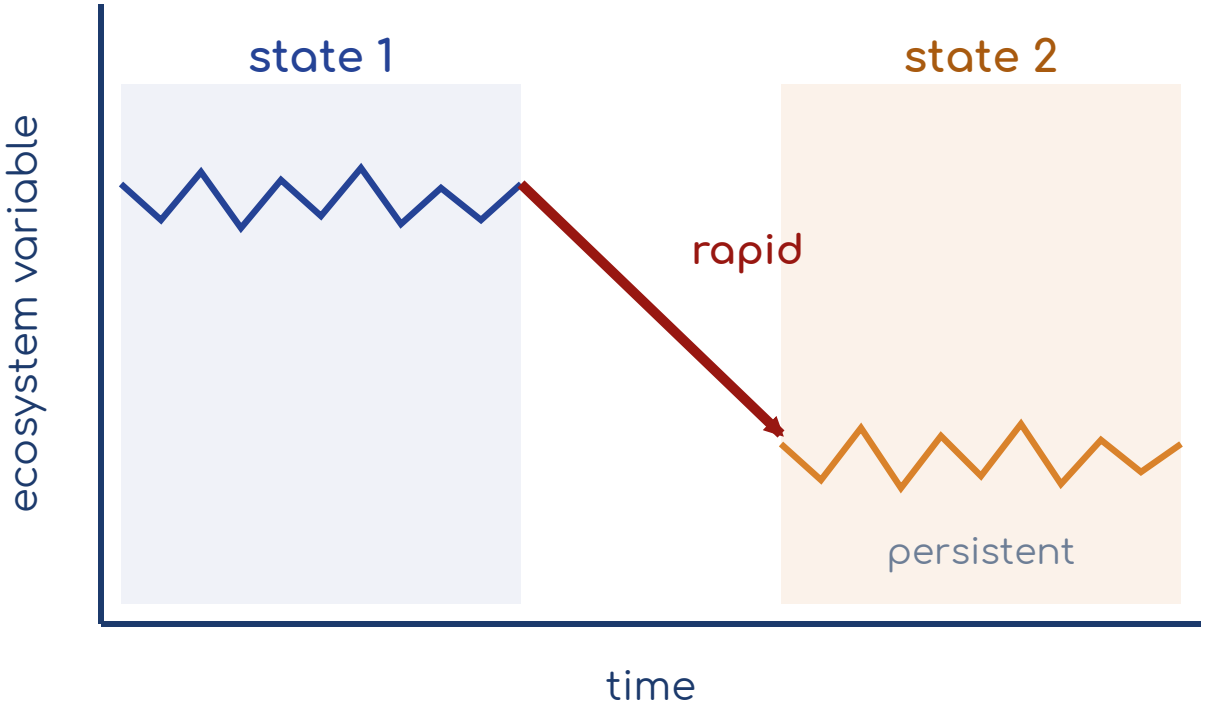
Candidate predictors: all other guilds, temperature, phosphate, silica, inorganic N and whitefish, each at lags 0–12 months

In-sample versus out-of-sample R^2





What is a regime shift?



- Variable **jumps** rapidly
- Faster than **background** change
- **New state** persists
- Found by **change points**
- **Drivers** not considered

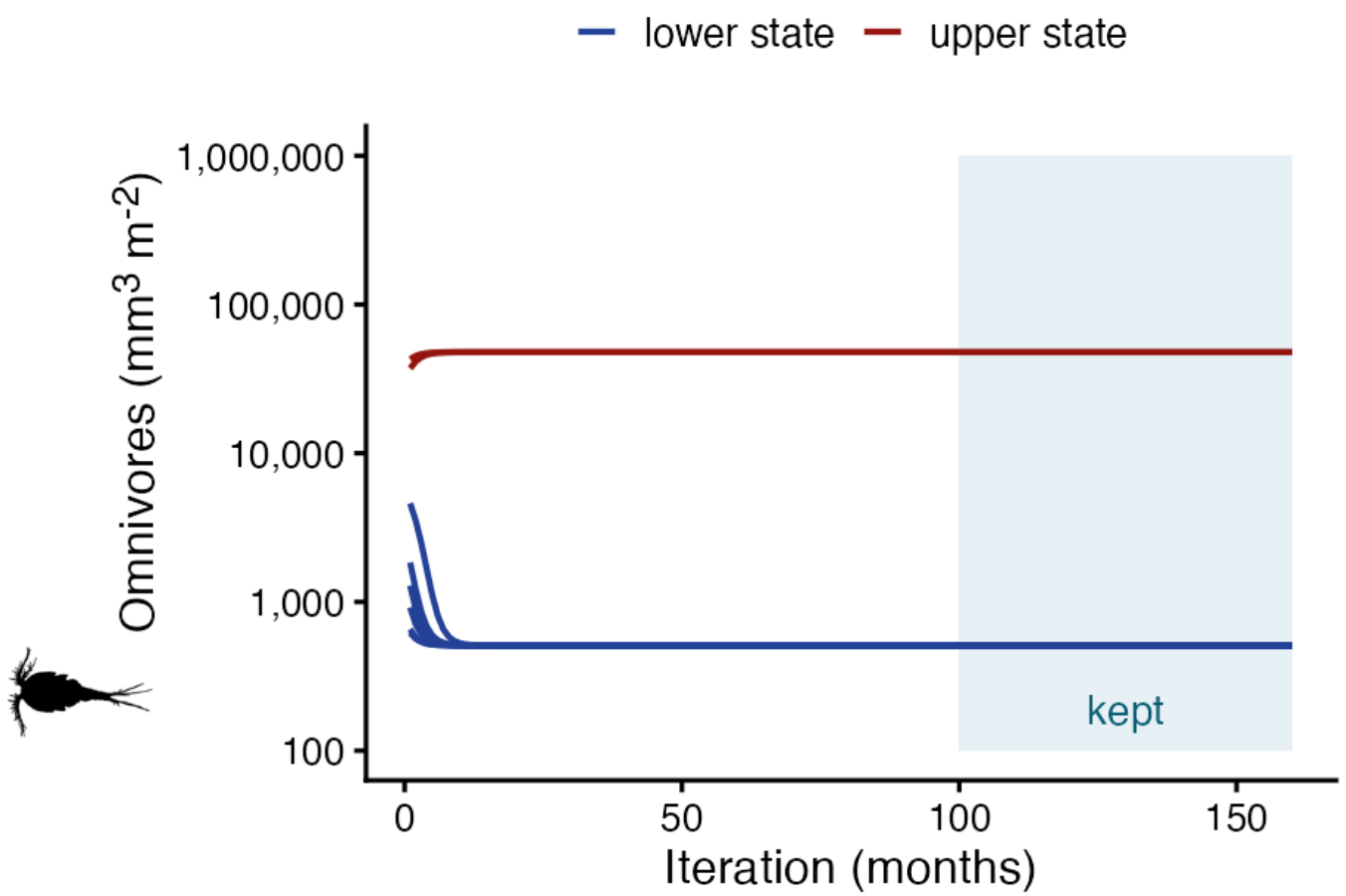
Ratajczak et al. 2018 · Folke et al. 2004

How the bifurcation diagram is made

1. **Fit GP-EDM**
   omnivore model
2. **Fix drivers**
   temperature, phosphate
3. **Pick driver level**
   green algae, silica
4. **Random start**
   two past months
5. **Predict next month**
   feed it back
6. **Repeat 160 times**
   keep last 60

7 Plot the kept values against the driver level

Iterating the model: one or two states?



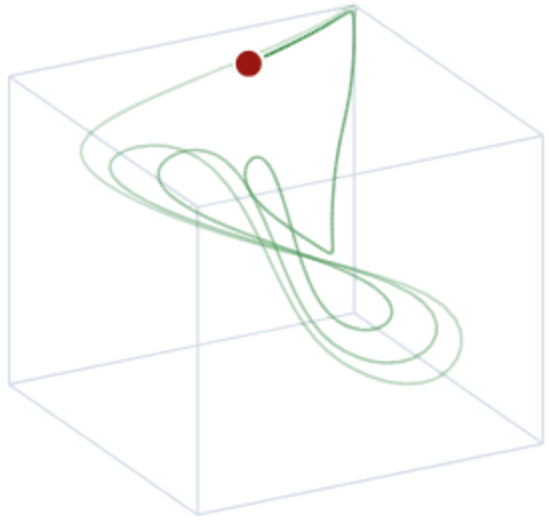
- Same drivers, different starts
- Starts reach **two** states
- Two states = bistability
- Example: small green algae 100 mm³ m⁻², high silica

GP-EDM: univariate embedding

Cyanobacteria (x)



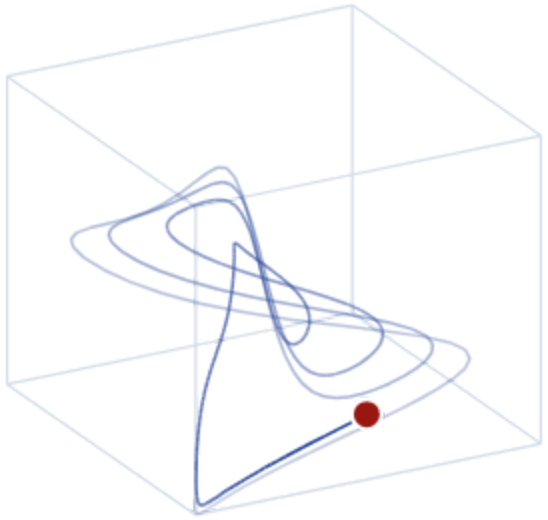
▼ lags of x only



Large herbivores (y)



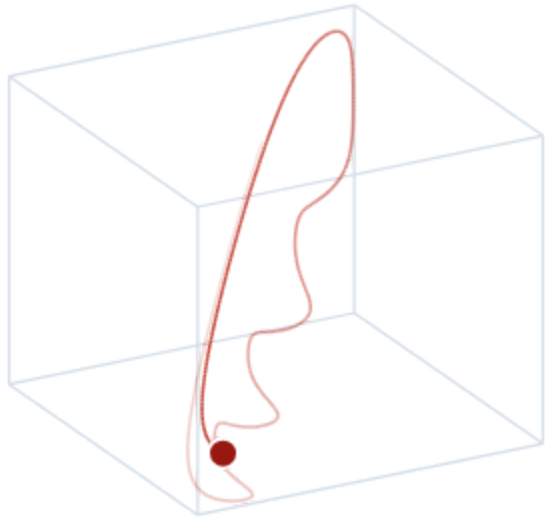
▼ lags of y only



Omnivores (z)



▼ lags of z only



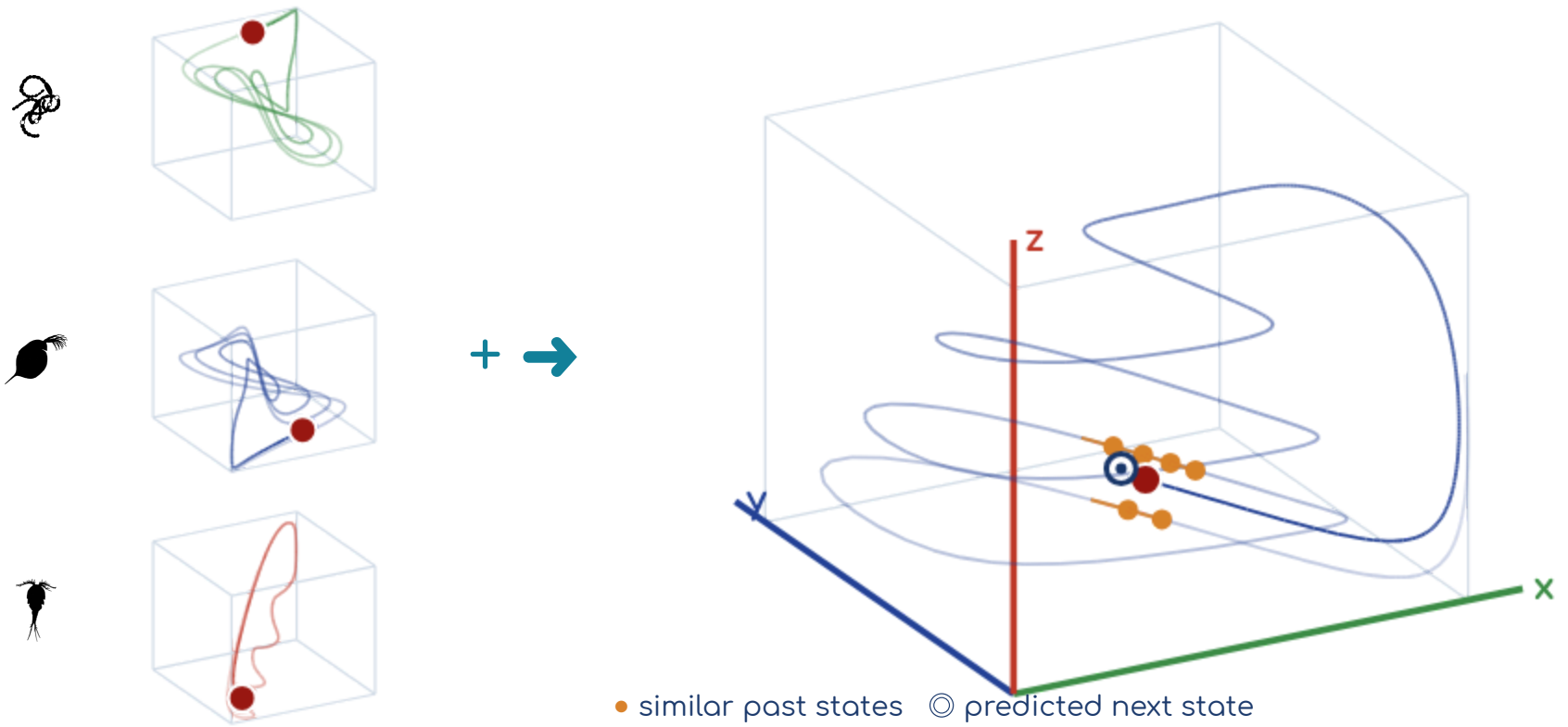
Embedding point
$(x_t, x_{t-\tau}, x_{t-2\tau})$

Main advantage of EDM
No model equations

Takens' theorem
Lags rebuild the attractor

Simulated three-species food chain; design after Ye & Sugihara 2016

GP-EDM: multivariate embedding



All species + lags

One GP model

ARD weights inputs

Dynamics

$x_{t+1} = f(x_t, y_t, z_t, x_{t-1}, y_{t-1}, z_{t-1}, \dots) + \varepsilon_t$

GP kernel

$k = \sigma^2 \exp(- \sum_i \phi_i (u_i - u'_i)^2)$

Simulated three-species food chain; design after Ye & Sugihara 2016

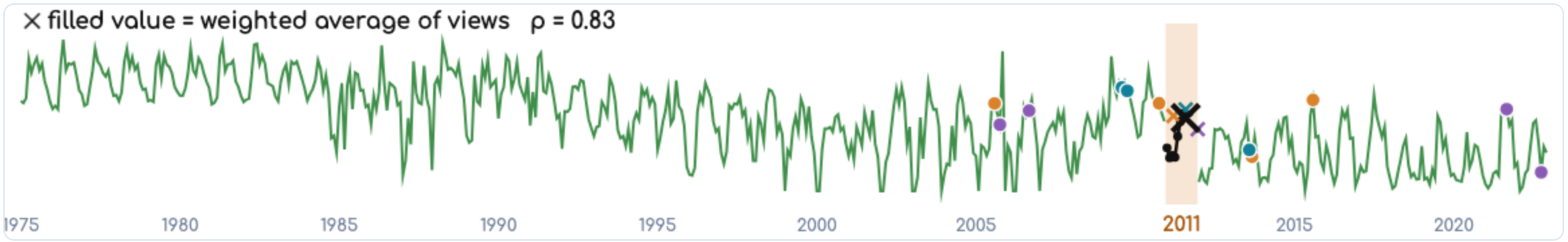
Multiview EDM: the core idea

Target: small green algae, not sampled in 2011 — zooplankton and abiotic variables give the views

View 1 t, t−2 t−1 — ρ = 0.82

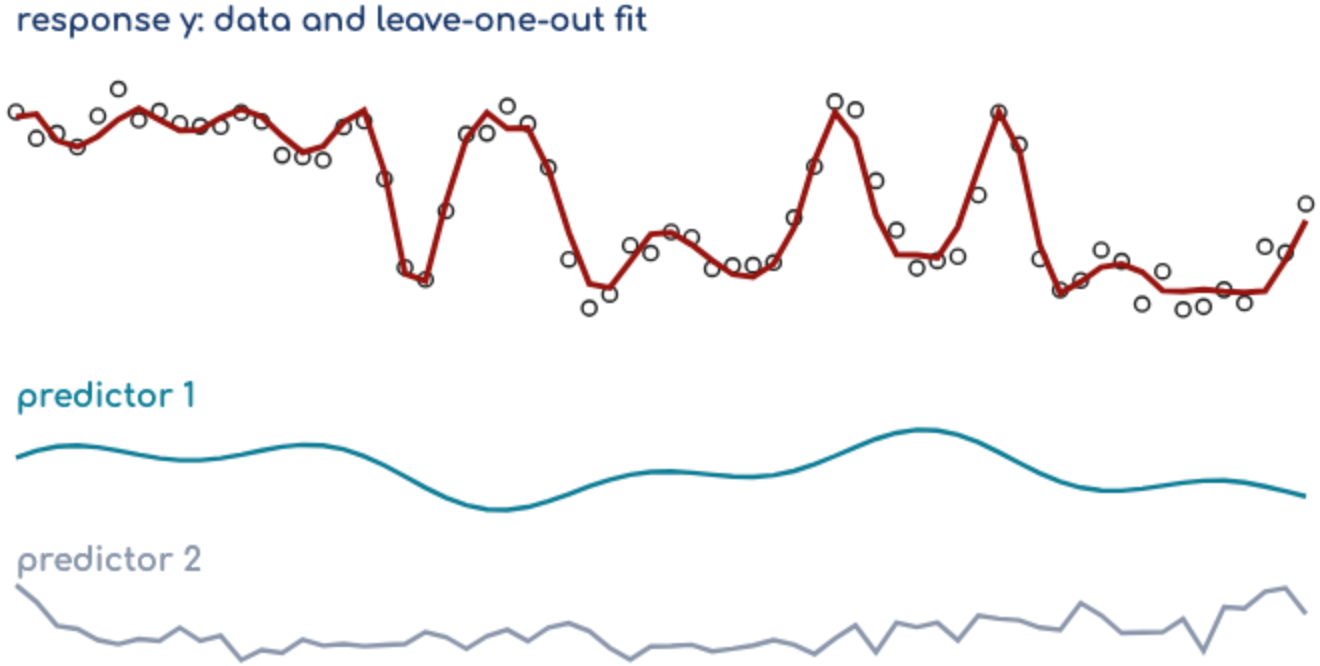
View 2 t, t−1 t−3 — ρ = 0.79

View 3 t, t−3 t−2 — ρ = 0.76



Real data, Lake Constance (IGKB). Grey: past states · red: month in 2011 · coloured: similar states · design after Carpenter et al. 2025 (Fig. 1)

Automatic relevance determination (ARD)





One ϕ per predictor

$k(x,x') = \sigma^2 \exp(- \phi_1(x_1 - x'_1)^2 - \phi_2(x_2 - x'_2)^2)$

Why drop predictors?

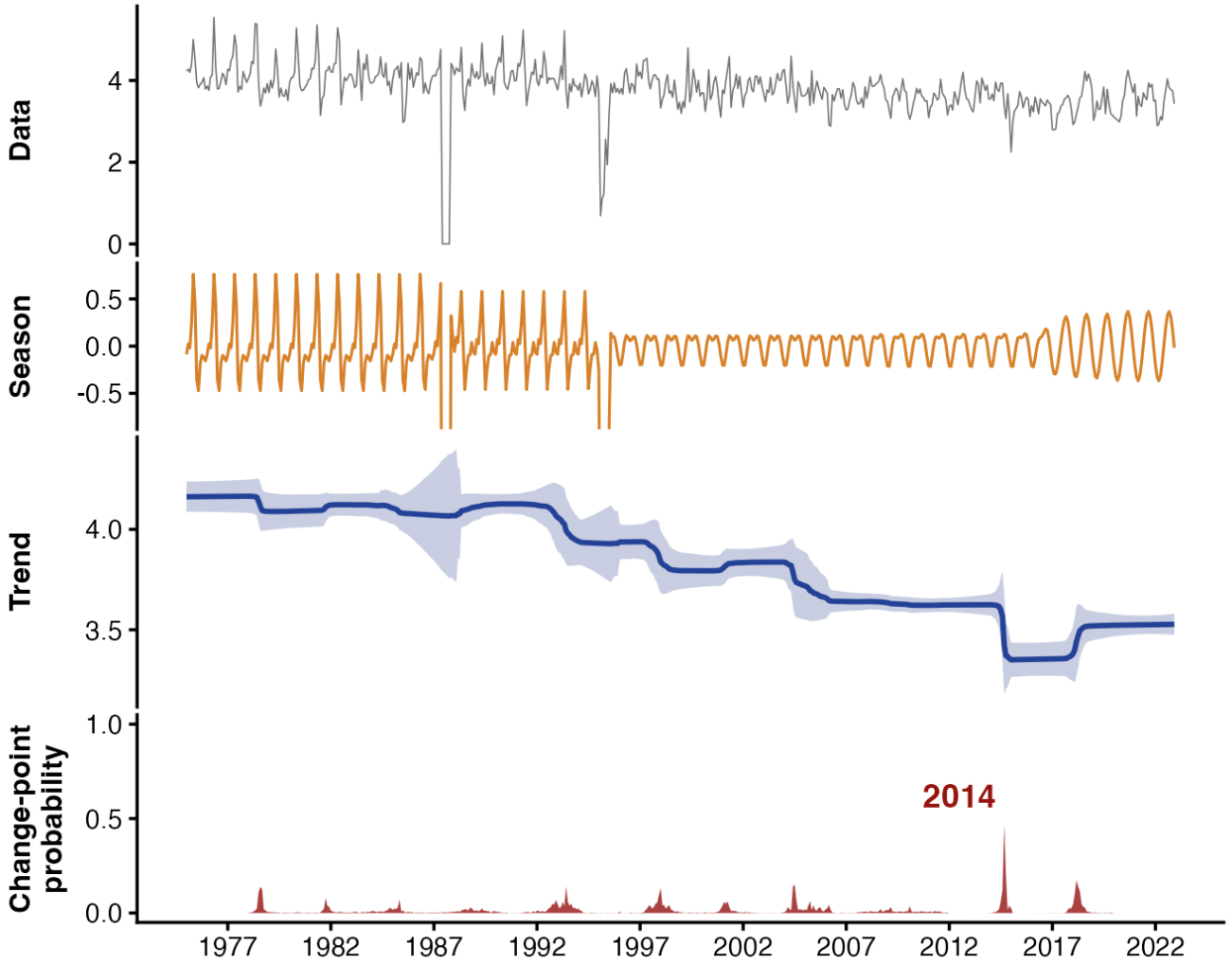
$\phi_2 = 0 \Rightarrow x_2$ never changes k

$\Rightarrow$ less noise, better LOO fit

Rule

$\phi_i \leq 0.001$: dropped

What does BEAST do?



Splits the series

y_t = trend + season + noise

Finds change points

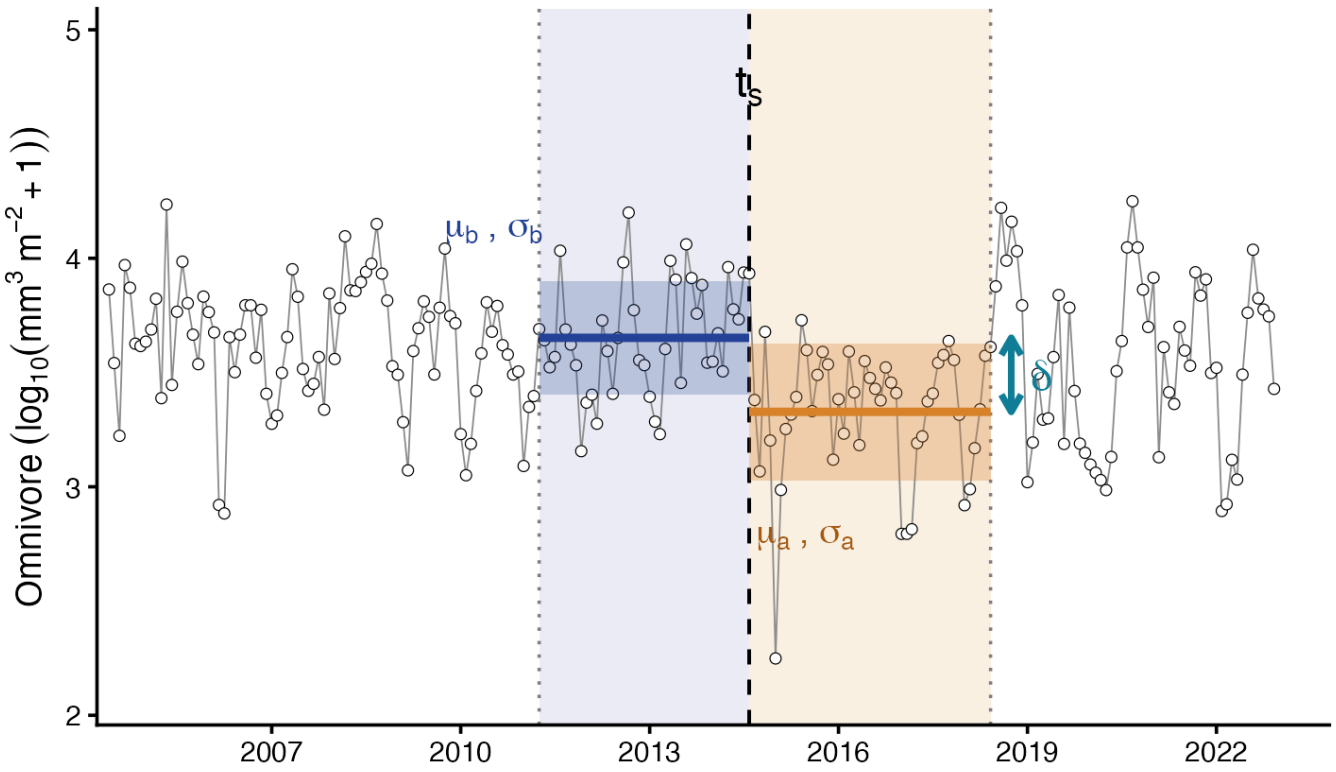
jumps of the **trend**

Gives a probability

Pr(change point) per month

Illustration: BEAST on the omnivores alone

Detecting regime shifts: criteria



1 Difference of the two chunks

$\delta = \mu_a - \mu_b \quad 0 \notin CI_{95\%}(\delta)$

CI = [0.19, 0.42] ✓

2 Abruptness: jump vs. usual swings

$$a = \frac{|\mu_a - \mu_b|}{\frac{1}{2}(\sigma_b + \sigma_a)} > 1$$

a = 1.19 ✓

Kept if 1 and 2 hold

Which genera? Zooplankton and mixotrophs



Invertebrate predators

Pr

Leptodora	73%
Bythotrephes	27%



Omnivores

Om

Cyclops	80%
Mesocyclops	17%
Acanthocyclops	1%
Thermocyclops	1%
Asplanchna	0%

+ 4 more



Large herbivores

He

Daphnia	91%
Eudiaptomus	6%
Bosmina	2%
Diaphanosoma	1%
Acanthodiaptomus	0%

+ 6 more



Mixotrophic flagellates

Mi

Ceratium	46%
Gymnodinium	45%
Peridinium	7%
other Flagellaten	1%
other Phytoflagellaten	0%

+ 14 more

Top genera by share of biovolume, IGKB 1975–2022

Which genera? Phytoplankton (1/2)



Cyanobacteria

Cy colonial

Anabaena	44%
Aphanizomenon	16%
Planktothrix	9%
Cyanothece	7%
Aphanothece	6%

+ 25 more



Green algae, small

Gr1 ≤ 550 µm³

Volvocales	30%
unidentified algae	24%
Chlorella	21%
Oocystis	7%
Characium	3%

+ 21 more

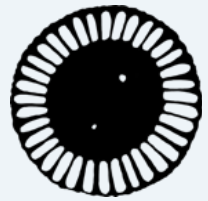


Green algae, large

Gr2 > 550 µm³ or colonial

other Chlorococcales	23%
Pandorina	18%
Mougeotia	16%
Chlamydomonas	10%
Ulothrix	6%

+ 46 more

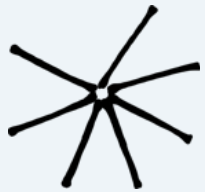


Diatoms, small

Di1 ≤ 800 µm³

Stephanodiscus	81%
Cyclotella	17%
Achnanthes	1%
Fragilaria	1%
Nitzschia	0%

+ 9 more



Diatoms, large

Di2 > 800 µm³ or colonial

Fragilaria	50%
Stephanodiscus	15%
Asterionella	14%
Cyclotella	10%
Tabellaria	4%

+ 18 more

Top genera by share of biovolume, IGKB 1975–2022

Which genera? Phytoplankton (2/2)



Gold algae, small

Go1 ≤ 250 µm³

Erkenia	63%
Chrysochromulina	23%
Chromulina	4%
Ochromonas	2%
Pseudokephyrion	2%

+ 13 more



Gold algae, large

Go2 > 250 µm³ or colonial

Dinobryon	42%
Mallomonas	17%
Uroglena	12%
Chrysophyceae	12%
Pseudopedinella	6%

+ 9 more



Cryptophytes, small

Cr1 ≤ 700 µm³

Rhodomonas	99%
Plagioselmis	1%
Cyathomonas	0%
Cryptaulax	0%
Chilomonas	0%

+ 1 more



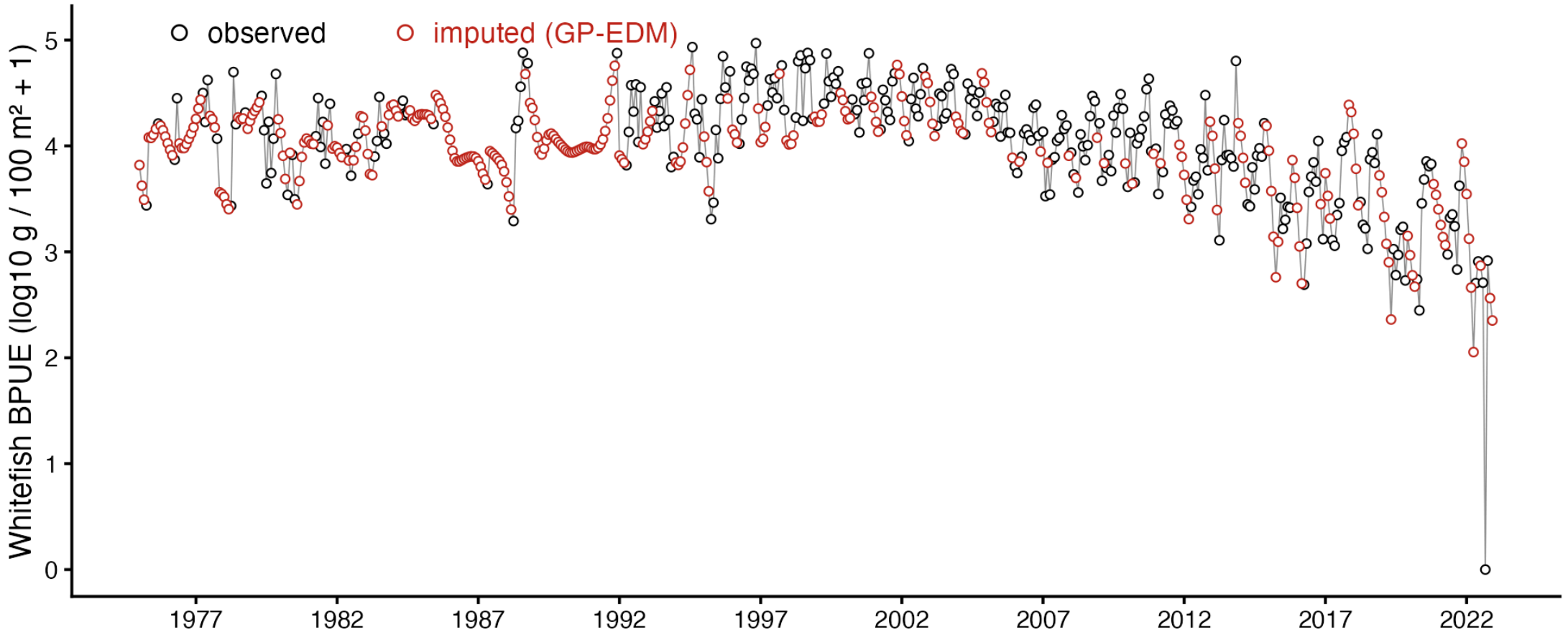
Cryptophytes, large

Cr2 > 700 µm³

Cryptomonas	100%
other Cryptophyceen	0%

Top genera by share of biovolume, IGKB 1975–2022

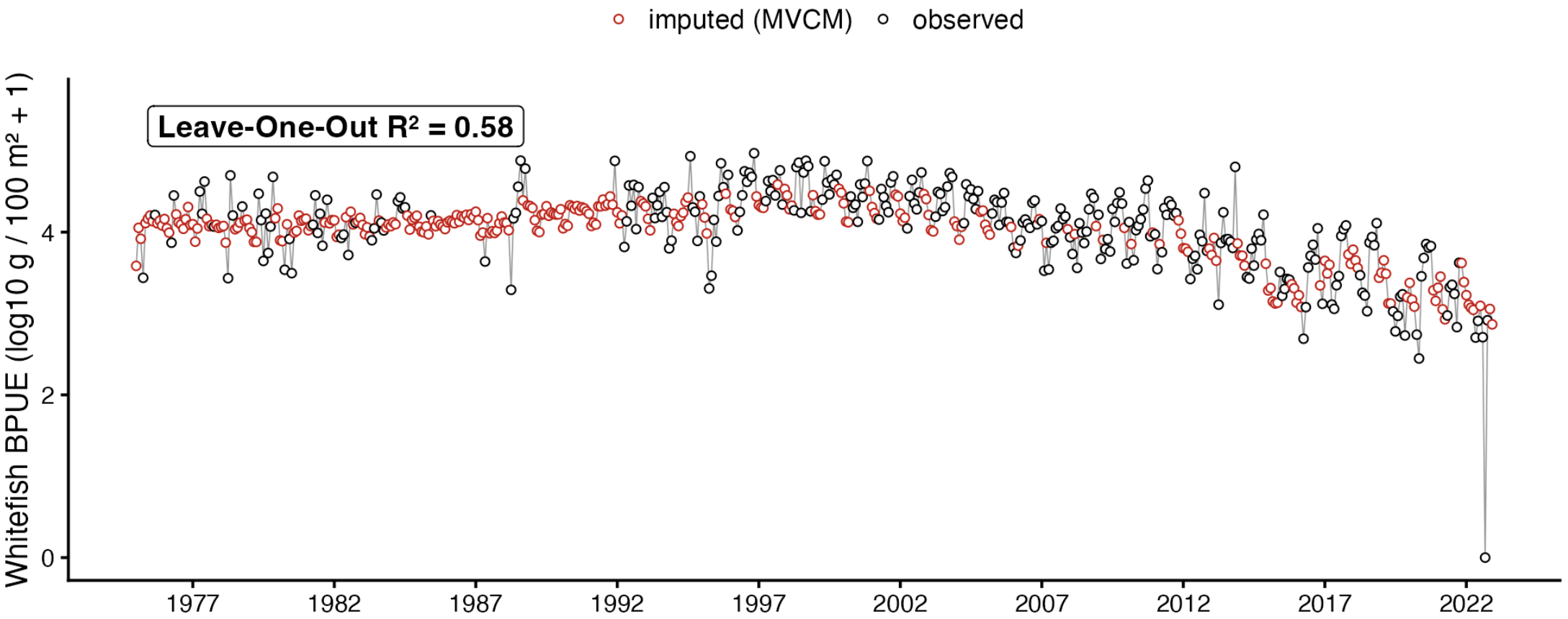
Whitefish time series: observed and imputed



287 of 576 months imputed (50 %)

GP-EDM, LOO R^2 = **0.55**

Whitefish time series: imputed with MVCM



287 of 576 months imputed

MVCM: Carpenter et al. 2025

GP-EDM version: Leave-One-Out R² = **0.55**

MVCM: multiview cross-mapping

1 Make views

4 inputs each, lags −2 to +2

2 Cross-map

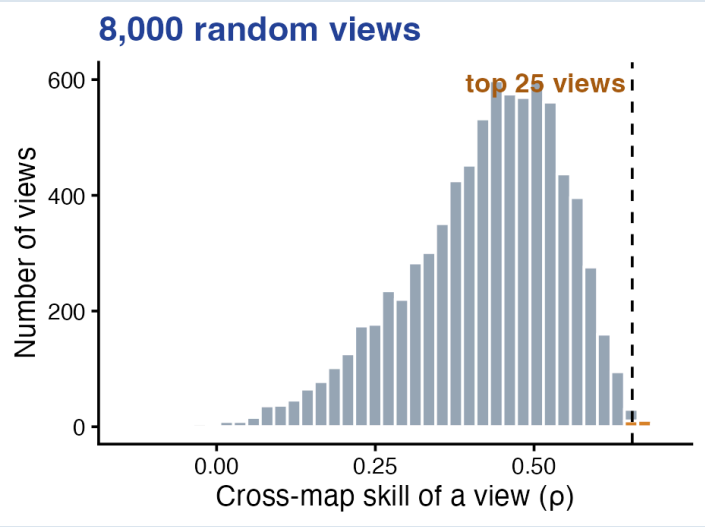
5 similar months, weighted

3 Rank views

by skill ρ

4 Average top 25

weights ∝ ρ



$\hat{y}_t = \sum_v w_v \hat{y}_{v,t} \quad w_v = \rho_v / \sum \rho$

8,000 random views

Simplex variant of MVCM

Carpenter et al. 2025; Ye & Sugihara 2016

References (1/2)

- Carpenter D, Deyle E, Park J, Saberski E, Sugihara G. 2025. Repairing gaps in ecological time series. Methods in Ecology and Evolution 16: 573–583.
- Eyring S, Merz E, Reyes M, Ntetsika P, et al. 2025. Distinct phytoplankton size classes respond differently to biotic and abiotic factors. ISME Communications 5: ycae148.
- Folke C, Carpenter S, Walker B, et al. 2004. Regime shifts, resilience, and biodiversity in ecosystem management. Annual Review of Ecology, Evolution, and Systematics 35: 557–581.
- Gilarranz LJ, Narwani A, Odermatt D, Siber R, Dakos V. 2022. Regime shifts, trends, and variability of lake productivity at a global scale. PNAS 119: e2116413119.
- Medeiros LP, Sorenson DK, Johnson BJ, Palkovacs E, Munch SB. 2025. Revealing unseen dynamical regimes of ecosystems from population time-series data. PNAS 122: e2416637122.
- Merz E, Saberski E, Gilarranz LJ, Isles PDF, et al. 2023. Disruption of ecological networks in lakes by climate change and nutrient fluctuations. Nature Climate Change 13: 389–396.
- Munch SB, Poynor V, Arriaza JL. 2017. Circumventing structural uncertainty: a Bayesian perspective on nonlinear forecasting for ecology. Ecological Complexity 32: 134–143.

References (2/2)

- Ntetsika P, Eyring S, Merz E, Reyes M, et al. 2026. Biotic interactions shape the realized niche of toxic cyanobacteria. Nature Ecology & Evolution 10: 1434–1448.
- Ratajczak Z, Carpenter SR, Ives AR, et al. 2018. Abrupt change in ecological systems: inference and diagnosis. Trends in Ecology & Evolution 33: 513–526.
- Straile D. 2025. Lake Constance. In: chapter in an Oxford University Press volume, https://academic.oup.com/book/61459/chapter/535428525 (title and editors to be completed).
- Sugihara G, May RM. 1990. Nonlinear forecasting as a way of distinguishing chaos from measurement error in time series. Nature 344: 734–741.
- Takens F. 1981. Detecting strange attractors in turbulence. Lecture Notes in Mathematics 898: 366–381.
- To S-W, Chakraborty S, Acevedo-Trejos E, Pomati F, et al. 2026. Future warming reshapes inorganic nutrient and plankton dynamics in a temperate lake. Limnology and Oceanography 71: e70447.
- Ye H, Sugihara G. 2016. Information leverage in interconnected ecosystems: overcoming the curse of dimensionality. Science 353: 922–925.
- Zhao K, Wulder MA, Hu T, et al. 2019. Detecting change-point, trend, and seasonality in satellite time series data to track abrupt changes and nonlinear dynamics: a Bayesian ensemble algorithm. Remote Sensing of Environment 232: 111181.

Map in slide 6 (panel a): Straile 2025. Panels b–d: monthly IGKB (phosphate, temperature) and IBKF (whitefish) data, 1975–2022.